DESIGN AND DECODING OF AN M-ARRAY PATTERN FOR LOW-COST STRUCTURED LIGHT 3D RECONSTRUCTION SYSTEMS

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ABSTRACT

3D shape reconstruction is one of the most important topics in computer vision due to its wide field of application. Among various technologies, structured light is considered to be one of the most reliable techniques. This paper addresses the problem of finding correspondences in a structured-light 3D shape reconstruction system. The work is based on the 3-symbol binary M-array encoded pattern proposed by Albitar. A new 6-symbol M-array pattern is designed with guaranteed minimum Hamming distance of three among all 3×3 windows. Improvements to the decoding algorithm are made, which allows successful identification of most symbols, and which corrects one possible error or missing symbol in each 3×3 window. Finally, refinements are made to the algorithm for finding correspondences, including back-projection of the captured structured light pattern to the projector plane to eliminate projective distortion.

***Index Terms*—** Structured light, correspondences, binary codes, feature extraction, low-cost system

1. INTRODUCTION

3D object reconstruction is becoming an increasingly important research area in both computer vision and image processing, due to its wide application in various fields. Examples include industrial part modeling and inspection, robot navigation, and 3D map-building. Depending on whether the sensing system needs to touch the object or not, there are two types of 3D capture technologies: contact methods and non-contact methods. Furthermore, as a large group of 3D capture techniques, non-contact methods involve two categories: non-contact active methods and non-contact passive methods. The goal of this paper is to accurately capture the 3D shape and color information of everyday small objects and documents at high speed, while using low-cost hardware, such as mobile phone cameras.

Time-of-flight systems are well known for their video rate processing speed. However, their low resolution and the random noise of the sensor have impeded their application for everyday small objects. Recently, an algorithm was proposed to obtain 3D scans of reasonable quality with a sensor that produces low quality data [1]. This algorithm is based on a new combination of a 3D super-resolution method with a probabilistic scan alignment approach that explicitly takes into account the noise characteristics of the sensor. However, the algorithm fails to capture accurate data for certain surface materials, such as highly specular objects. Currently, the super-resolution takes almost 95% of the current run-time, which is too much for low-cost consumer applications.

Another large group of 3D-capture techniques is based on triangulation [2]. Examples include stereoscopic systems, laser scanning systems, and structured light systems [3]. Structured light systems are based on projecting a light pattern and capturing the illuminated scene from one or more points of view. To accurately obtain 3D shape information, controlled illumination is highly preferred [4]. By adopting such a system, fewer frames, even as few as one frame for each viewpoint, are processed to obtain the range map. Low-cost hardware and fast processing become possible [2].

Pattern encoding and decoding remain major problems for structured light systems. Time-varying pattern encoding techniques have the highest measurement accuracy [5]. However, the apparatus needed for time-varying pattern encoding is too expensive and complex for low-cost consumer applications. On the other hand, spatial neighborhood codification strategies are attractive for use in low-cost systems, because the 3D information can be reconstructed from just one projected pattern. Such structured light patterns may use gray-scale, color, or shape to encode information about the spatial location. However, spatial variations in ambient illumination and object color may reduce the accuracy with which the structured light pattern is rendered and captured. Therefore, our research focuses on binary patterns using spatial neighborhood codification strategies.

Several binary patterns using spatial neighborhood codification strategy have been proposed. The pattern proposed by Vuylsteke et al is based on De Bruijn sequences [6]. Maruyama proposed a binary structured light pattern of mul-

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tiple slits with random cuts [7]. Malzbender et al recently proposed another binary pattern coded with a special M-sequence using two primitives: solid lines and dashed line segments [8]. For this method, the projector and the camera must be carefully aligned such that for each primitive detected in the captured image, given its corresponding column in the projector's image plane, the corresponding row can be determined accordingly. Thus the two-dimensional matching problem can be reduced to a one-dimensional problem. The special one-dimensional M-sequence can correct a 1-bit error in every 11-bit block. However, perfect alignment remains a challenging problem. The decoding process is not robust enough due to easily handle missing line segments.

Finally, the pattern proposed by Albitar is very suitable for low-cost consumer applications [9]. Based on the theory of M-arrays, a pattern of 27 rows and 29 columns is generated with a 3×3 window property [10]. Three basic shapes: dash, disc, and annulus, are chosen to be the primitives. Using the method of fill-and-test-as-you-go-along, M-arrays of any size and a desired window property can be constructed given a sufficient number of primitives [11]. The minimum Hamming distance in the M-array pattern with a $v \times w$ window property is defined to be the minimum number of different digits at any given pixel location among all overlapping $v \times w$ windows [12]. By definition of the M-array, all M-arrays should have at least a minimum Hamming distance of one. In our correspondence finding problem, if there are no errors nor missing symbols, a minimum Hamming distance of one is sufficient to find correspondences between captured images and the original pattern. In practice, errors and missing symbols can easily occur. To correct t errors or missing symbols within a codeword, a minimum Hamming distance of $(2t + 1)$ is required [11].

In this paper, a newly designed M-array pattern with a minimum Hamming distance of three is constructed using the fill-and-test-as-you-go-along method mentioned above. Since the native aspect ratio of our hardware is 4:3 (horizontal:vertical), the new pattern contains 27 rows and 36 columns. It satisfies a 3×3 window property. In this case, three primitives are far from sufficient to satisfy the above constrains for decodability. So we use six primitives in the pattern: the three previously proposed primitives – dash, disc, and annulus, and three new primitives – triangle, star, and diamond that we have introduced. Given six primitives, an M-array pattern of 33 rows and 44 columns can be constructed with the same minimum Hamming distance of three as the 27×36 pattern. We also make improvements to the decoding algorithm. We are able to label each symbol to one of the six classes. The four neighbors for each symbol are found to get the codeword for its surrounding 3×3 window. We implement an advanced matching scheme with one digit error correction in the processing pipeline. Moreover, we propose a refinement stage based on projecting the camera scene back to the projector's image plane. With this back-projection refinement, we obtain an increase in the symbol matching rate.

The rest of this paper is organized as follows. Section 2 describes the detailed procedure for constructing the new pattern. The decoding algorithm is illustrated in Sec. 3, followed by the refinement procedure to the matching result in Sec. 4. Finally, the experimental results and conclusions are shown in Secs. 5 and 6.

2. STRUCTURED LIGHT PATTERN ENCODING

The fill-and-test-as-you-go-along method is used to construct an M-array pattern [11]. It first randomly generates a $v \times w$ window from the alphabet of primitives. Then, the window is moved one column to the right; and three primitives are chosen that satisfy the minimum Hamming distance property (three in our case), compared to the first window. The process is repeated to fill the first v rows. Then, the first w columns are completed by moving down one row at a time. Next, one symbol is added each time at the lower right corner of a window, until the pattern is filled. If at some point, adding any combination of the primitives cannot maintain the $v \times w$ window property, we start all over again from the beginning.

Using this procedure, six primitives are required to generate a 27×36 pattern with a minimum Hamming distance of three. The primitives we used are shown in Fig. 1.



Fig. 1. Primitives used in the newly designed pattern. (a) Dash; (b) Disc; (c) Annulus; (d) Triangle; (e) Star; (f) Diamond.

The final design is shown in Fig. 2(a). In Figs. 2(b) and 2(c), we compare the Hamming distance histogram of the newly designed pattern and that of the pattern proposed by Albitar. It can be observed that the minimum Hamming distance is increased from 1 to 3 in the newly designed pattern [13]. The average Hamming distance is also increased from 6.0 to 7.5.

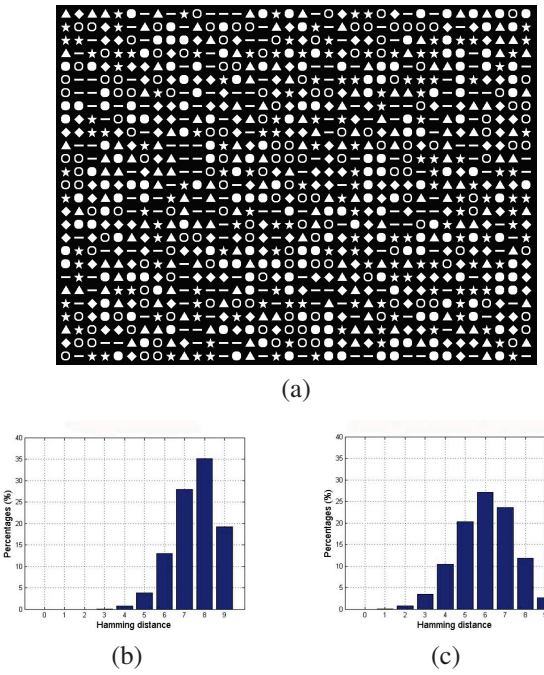
3. STRUCTURED LIGHT PATTERN DECODING

There are four major stages in the decoding algorithm, including preprocessing, classifying the detected symbols, neighborhood finding, and matching.

3.1. Preprocessing

The color image is first converted to gray scale, then Otsu's threshold is used to get a binary image [14]. All the connected

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(a) (b) (c)

Fig. 2. (a) Newly designed M-array pattern with six primitives (27 rows and 36 columns). (b)-(c) The Hamming distances histogram for the newly designed M-array pattern and Albitar's proposed pattern, respectively.

components in the binary image are found. To remove potential noise, we ignore a connected component if the number of its pixels is smaller than 60.

3.2. Classifying the detected symbols

A two-step procedure is used to classify the symbols. It is determined whether or not each symbol belongs to one of the three previously proposed classes first, i.e. dash, disc, and annulus, based on the first five shape descriptors described below. If the symbol does not belong to any of them, the sixth shape descriptor symmetry is used to test it for the three new classes we introduced – triangle, star, and diamond. All shape descriptors we used are described below.

1. Number of edges. The number of edges N_{edge} in each connected component can help us find the annulus, which is the only primitive with two edges.

2. Area. A loose threshold range is assigned to the area of each class based on experimental results, in case several symbols are fused together and should not belong to any class.

3. Distances from the centroid to edge pixels. The distances between the centroid of each symbol and all points on the edge, which is extracted by the Sobel operator, are calculated. The ratio between the maximum $D_{\max}$ and minimum $D_{\min}$ is used as an important descriptor to distinguish the dash and the disc [13]. In addition, we propose to use overlapping threshold ranges for the dash and the disc, because non-overlapping threshold ranges fail to work for different objects in the presence of projective distortion.

For the dash, the direction of the maximum distance between its centroid and its edge pixels is recorded as the search direction for the neighborhood finding stage.

4. Minimum circle. The minimum circle here is defined as the smallest circular region centered at the centroid of a symbol that would contain all pixels of that symbol. The radius of the minimum circle is $D_{\max}$. The area of the circle S_{circle} should be $\pi D_{\max}^2$. And the area of a symbol S_{symbol} is just the number of its pixels. It can be observed that the ratio $\frac{S_{\text{symbol}}}{S_{\text{circle}}}$ is guaranteed to be less than or equal to 1. Dashes have the smallest $\frac{S_{\text{symbol}}}{S_{\text{circle}}}$ value among all primitives, while discs have the largest value. The minimum circle for each primitive, and the corresponding ideal value of $\frac{S_{\text{symbol}}}{S_{\text{circle}}}$ is shown in Fig. 3.

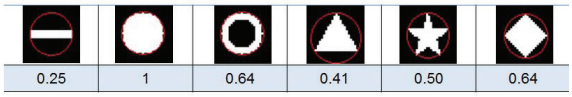


Fig. 3. All primitives with their minimum circles. The numbers are the ideal $\frac{S_{\text{symbol}}}{S_{\text{circle}}}$ value for each primitive, when no projective distortion exists.

5. Bounding box. The bounding box is defined as the rectangular region that just contains a symbol. The aspect ratio of the rectangle is an important feature to distinguish dashes from other classes. The ratio between the area of a symbol and the area of its bounding box is another feature.

6. Symmetry. For the three new classes that we have introduced, the symmetry property of the symbol is used. The diamond is the only class that has four-fold central symmetry. In addition, the triangle has the least top-bottom folding symmetry. To quantify the symmetry property of a symbol, we rotate it with respect to its centroid by 180°, and calculate the area S_{rotate} of the overlapped region before and after the rotation. The ratio $\frac{S_{\text{rotate}}}{S_{\text{symbol}}}$ is then used to test the symmetry of a symbol.

3.3. Neighborhood finding

It is known from the previous section that the dashes provide the search direction for finding their neighbors. We find the neighbors of all the dashes first, and propagate the direction information to their neighbors. We repeat the process until every symbol is examined for neighborhood finding. Both the norm and angle of the vector connecting one symbol and another are used to find its four neighbors. By setting an experimentally determined threshold on the norm, we find several candidates. The angle is then compared to the search direction of that symbol to find its upper, lower, left, and right neighbors.

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3.4. Error correction and matching

We keep Albitar's definition of the codeword of a window in this section [9]. The method of vote is used to find the correspondences between the camera frame and the projected pattern [11]. With the guaranteed minimum Hamming distance of three, we ignore one digit in the 3×3 window at a time, search for its correspondence in Fig. 2(a), and record one vote for the match. We repeat this for each of the nine digits in the window. For each symbol, the winner with the highest number of votes is claimed to be the true match.

This advanced matching procedure increases the matching rate. It also suffers from mismatched symbols. We set a minimum threshold on the number of votes of the winner; so that we can always get reasonable results.

4. REFINE THE POINT CLOUD

A refinement to the correspondence matching result is proposed. Once we have the point cloud reconstructed from the correspondences and system calibration results, the points can be transformed back to the projector's image plane, where the symbols do not suffer from projective distortion. The decoding task is much easier and more robust in this back-projected image. We propose to generate the full back-projected image through interpolation, and run the image through a simpler version of the decoding process described above. At last, the original matching result is refined with the matching result from the back-projected image.

The Shepard interpolation method is used with the power of the distance between each interpolated point and data point set to be 4 [15]. The decoding process is also simplified, given that there is almost no projective distortion. The range of the area of each symbol is more predictable. We can also assume the horizontal axis to be the search direction for the neighbors.

5. EXPERIMENTAL RESULTS

The system consists of one Logitech C950 Webcam[1] (1600 $\times$ 1200 pixels) and one AAXA M2 Micro Projector[2] (1024 $\times$ 768 pixels). The backing board itself (0mm), two pieces of folded paper of height 32mm, shown in Fig. 4, and 52mm are tested, respectively. Both Fig. 4 and Table 1 show that refinement improves the matching rate. The larger the height of the object in z direction, the higher the error rate is.

Table 1. Comparison of the error rate (%).

	0mm	32mm	52mm
Albitar's algorithm*	0.26	2.68	4.21
Our algorithm without refinement	0.31	1.13	3.40
Our algorithm with refinement	0.11	0.62	2.98

*Implemented by the authors based on [9].

[1]Logitech, Newark, CA 94560

[2]AAXA Technologies Inc., Tustin, CA 92780

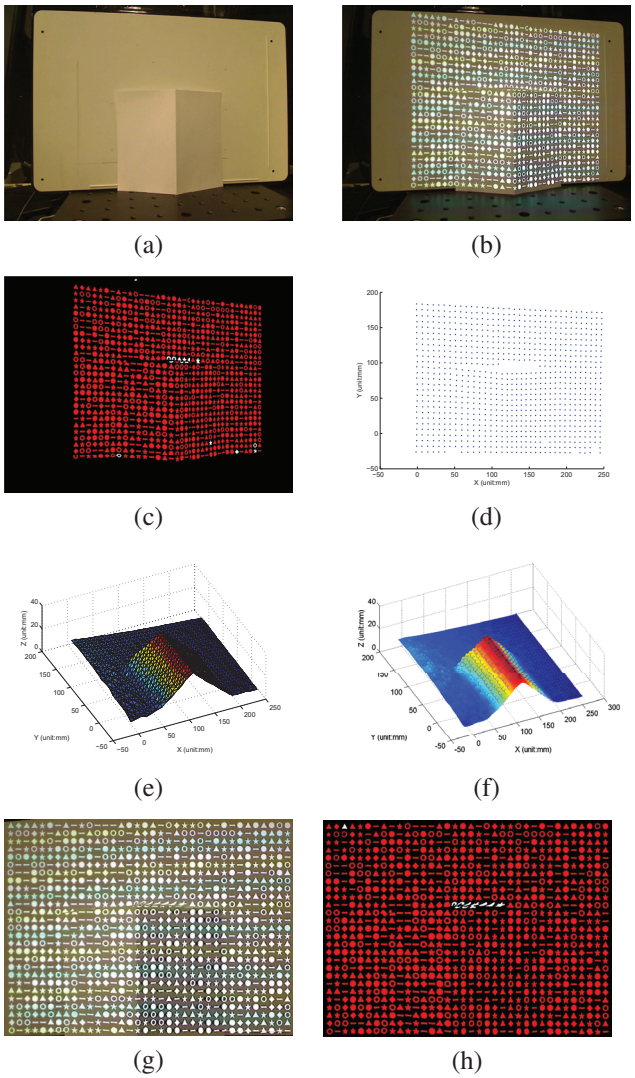


Fig. 4. (a) The original scene; (b) Captured image with the projected pattern; (c) Symbols correctly matched to the original pattern are marked in red; (d) Reconstructed point cloud; (e) Delaunay triangulation surface; (f) Shepard interpolation result; (g) The back-projected image; (h) Symbols correctly matched in the back-projected image are marked in red.

6. CONCLUSIONS

In this paper, we discussed several important 3D reconstruction methods that are relevant to low-cost consumer applications. A binary M-array pattern with six primitives was proposed. Improvements were made to decode the pattern correctly. With a guaranteed minimum Hamming distance of three, the new pattern has 1-bit error correction capability for each 3×3 window. The proposed back-projection procedure refines the correspondence finding result.

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